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PlayList : spring boot full course
channel : india free internship
youtube channel <https://www.youtube.com/@IndiaFreeInternship>
source Code: [spring boot source Code](#)

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SESSION 1

-
- 1. SPRING FRAMEWORK**
 - 2. SPRING BOOT**
 - 3. SRPING DATA JPA**
 - 4. SPRING WEB MVC**
 - 5. SPRING REST**
 - 6. SPRING SECURITY**
 - 7. MICROSERVICE**
 - 8. SPRING BATCH**
 - 9. SPRING SCHEDULAR**
- =====

SESSION 2

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Technolgies : webPage(servlets) + Database(JDBC/JPA) + Email (Java Mail) + Message(JMS)

Limitations:

- Start coding from line zero
- Development takes more time
- More chance of error
- Design pattern applied manually

Framework:

- Ready-made structure
- In-built technologies
- Design patterns already implemented

RAD => Rapid Application Development

- Fast coding
- Less Error

Spring Framework:

- used for Web Application or Distributed appliation
- It has own Apis
- It Follow Container System

Spring Container

-
- Find/Scan Classes[spring bean]
 - Create Object
 - provide DATA
 - Link Object based on Relation
 - Finally Destroy

Depedency Injection (DI)

Depedency Injection is a concept/theory which is implemented by spring IoC (Inversion of control) also called as spring container.

Theory	programming
oops	core java
ORM	HIbernate with JPA
DI	Spring IoC(Inversion of control)

session 3

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Application

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3 Type of application

- (i) Web Application**
- (ii) Distributed Application**
- (iii) Microservices Application**

(i) Web Application : Collection of web pages run under webserver.

Application : - n services

Project :- n Module

eg : Gmail Application

User(Register and Login) , Inbox, Sent, Drfats, Setting, etc

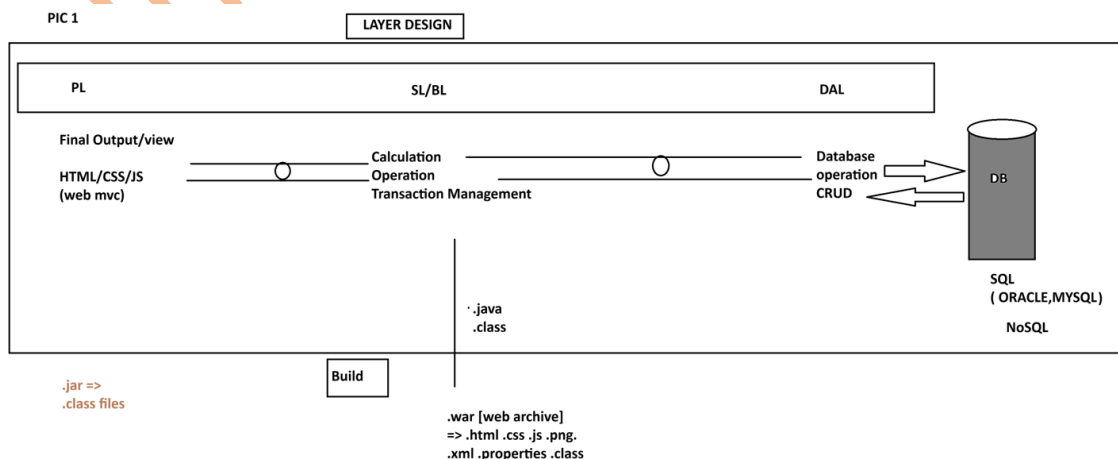
Amazon Application

User(Register and Login) , search, Cart, Payment, ...etc

=> Implementation of service / module: - This is implemented using 3 Layers Design.

- 1. DAL : Data Access Layer (Database Operations)**
- 2. SL : Service Layer (Calculation/Operations/txn mngmt)**
- 3. PL : Presentation layer (view /Dynamic web pages/MVC)**

PIC 1



(ii) **Distributed Application** : An Application that runs at mutiple devices is called as distributed appliation.

xyz Bank application

This appliation can be accessed in different ways. Example.

eg:

1. Manually Process
2. NetBanking (Web App)
3. Mobile Banking
4. IVR
5. ATM / DC
6. 3rd party app (Gpay, PayPhone..etc)

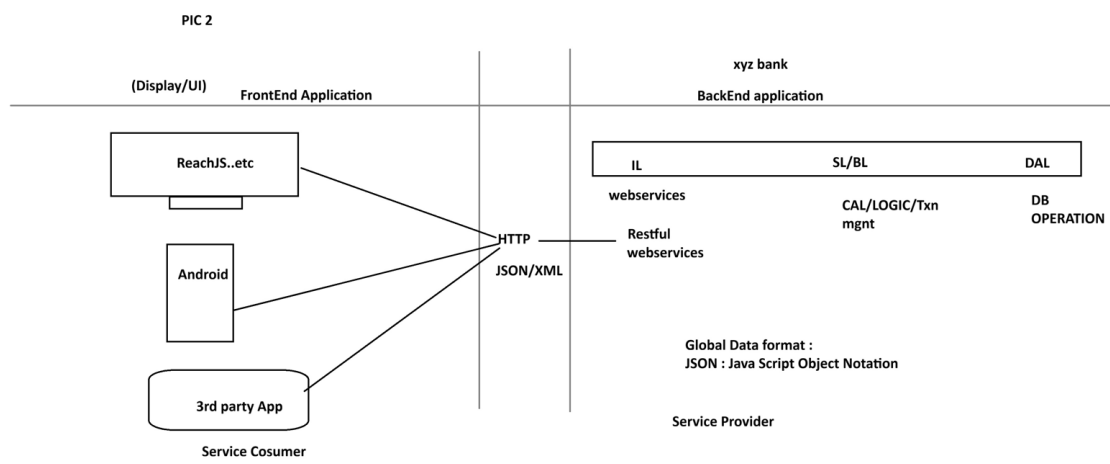
.....

-But here business logics exist at backend application.

- Fontend applications are implemented using Angular, Android...etc

- App connected using HTTP Protocol using Global Data Format (JSON/XML)

PIC 2

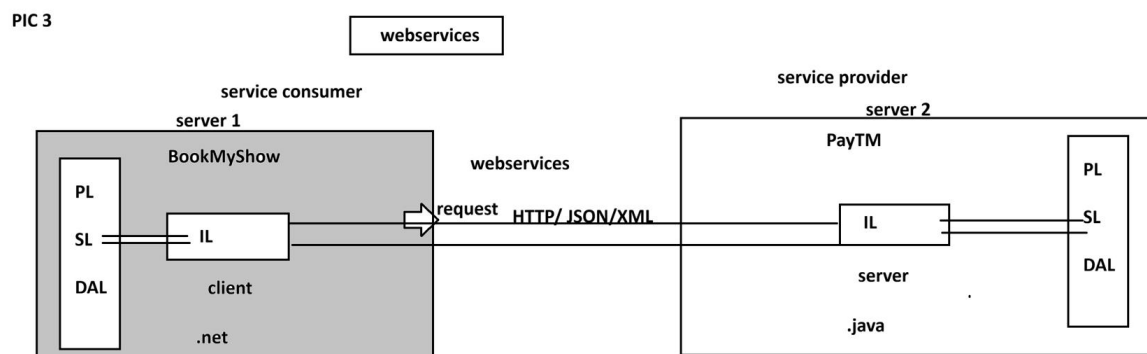


WebServices:

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Intergration of multiple (at least two) applications which runs on different servers and implemented using different language (java, .net, php, python....etc)

PIC 3



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session 4

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SPRING FRAMEWORK

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CORE

- 1. DI and IOC**
- 2. Container Types**
- 3. Annotation Configuration**
- 4. @ComponentScan**
- 5. @Component**
- 6. @Value**
- 7. Java Configuration**
- 8. @Configuration**
- 9. @Bean**
- 10. @PropertySource**
- 11. Environment**
- 12. Lifecycle methods**
- 13. Runners**
- 14. @ConfigurationProperties**
- 15. Properties file**
- 16. YAML File**
- 17. Profiles**
- 18. Lombok API**

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spring core chapter 1

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Core:

This module/chapter gives all rules and guidelines to work with spring container.

- 1. DI and IOC**

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Spring Container :- It takes care of object.

- 1. Find/Scan our Classes(spring bean)**
- 2. Create Object**
- 3. Provide Data**
- 4. Link Object**
- 5. Destroy the Object**

=>For this programmer has to provide two inputs-

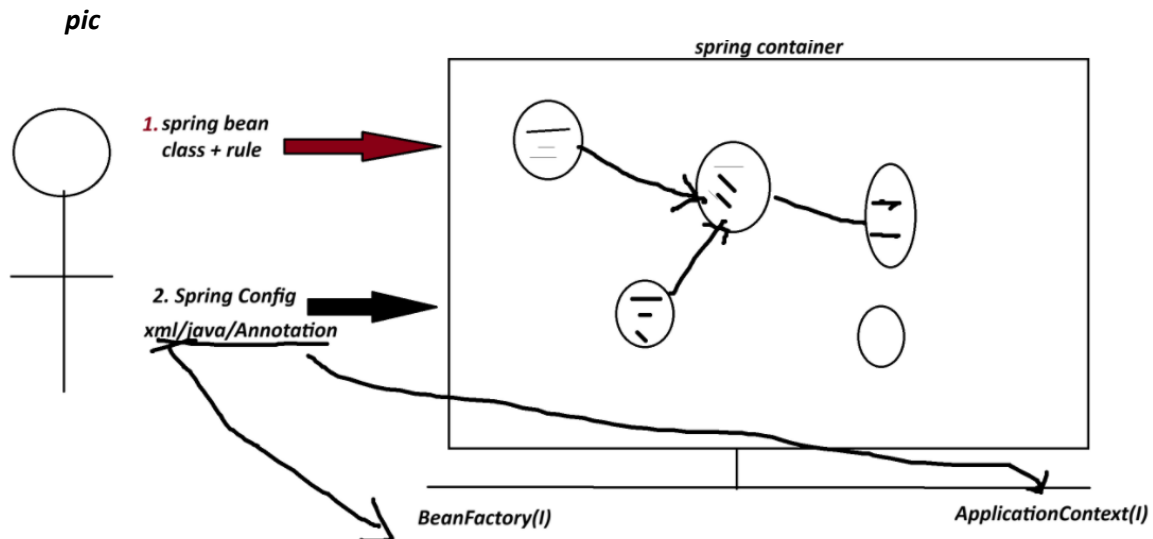
- 1. Spring Bean : class+rules given by spring container.**
- 2. Spring Configuration : This is main input contains details like object name, relation with objects, data....etc.**

xml/java/Annotation

**** Spring Container 2 Types:**

- 1. BeanFactory (Old container) : support only XML configuration file.**

2. **ApplicationContext (New container) :** support xml/java/Annotation configuration.



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Dependency Injection (DI)

Theory	programming
OOPs	java
ORM	Hibernate with JPA
DI	Spring IoC(Spring Container) -> Inversion of Control

=> **Depedency** :- An instance variable (Field) exist inside a class (spring bean)

1. **Primitive Type Dependency (PTD)** :- If a variable is created using (8+1) datatype.
(byte, short, int, long, float, double, boolean, char +string)
(Their wrapper classess too)

2. **Collection Type Dependency (CTD)** :- If a variable created using(4) types:
(java.util.package)
(List,Map,Set and Properties)

3. **Reference Type Dependency (RTD)** :- If a variable is creatd using other class/interface

Injection(4)

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=> It give/provide/assign existed provide data to Depedency(Inside object)

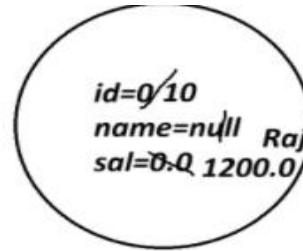
1. **Setter Injection (SI)** :- provide data to variable using its set method(setter).

2. **Constructor Injection(CI)**: provide data creating object using parameterized constructor.

3. **LookUpMethod Injection(LMI)** :- Spring bean Scope
[When parent bean is sigleton and child bean is prototype]

4. **Interface Injection(II)** : - Not Exist in Spring F/W.

```
class Employee{
    int id;
    String name;
    double sal;
}
```



Employee e=new Employee()

```
class Product{
    int id; //PTD
    String name; //PTD
    Boolean exist; //PTD
```

```
class Vendor{}
```

```
interface Service{}
```

```
List<String> models; //CTD
Map<String,Integer> data; //CTD
```

```
Vendor vob; //RTD
Service sob; //RTD
```

```
}
```

SESSION 5

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1. Spring Bean

2. STS Installation

Spring Bean

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-It is class that follow rules given by Spring Container.

Rule

- class must be public Type.
- Recomended to keep in a package (basePackage Rule)

ex :

com.app.service

com.app.controller

- Fields(Variables) are option, if required recomended to keep private.

```
private Integer empId;
```

```
private String empName;
```

- If variable present in class,

=>define default constructor with setter/getter

=>(or/and) Parameterized Constructor.

ex:

```
public class Employee{  
    private Integer id;  
    private String name;  
    public Employee(){  
    }  
    public void setId(Integer id){  
        this.id=id;  
    }  
  
    public Integer getId(){  
        return id;  
    }  
}
```

e. We can override Object(c) methods:

toString() equals() hashCode()

[because non-private, non-final,non-static]

f. Inheritance Rule:

Allowed ->Only Spring API is allowed (ie spring F/W classes and interface)

not allowed-> EJB, Servlet

g. Annotation Rule:

Only we can add Spring API Annotation and Inetegration API Annotations

(Hibernate with JPA, Restful webServices annotations);

ex:

@Component, @Service, @Repository.. etc.

STS Installation

=====

step 1:

go to official website

spring.io

link : - <https://spring.io/tools#eclipse>

step 2 download STS according your OS

step 3

Rigth click on ZIP file

click on "Extract Here"

step 4

Open extracted folder

Run : SpringToolsForEclipse

Final STEP STS installed successfully.

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1. Spring Bean

Simple Java class that follow rules given by container.

2. Spring Configuration File(XML)

It gives object

details (name, data, links with other objects ... etc).

3. Test class=> Just find, container created or not? object is created or not?

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Spring core XML Tags

=====

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

1. To create Object

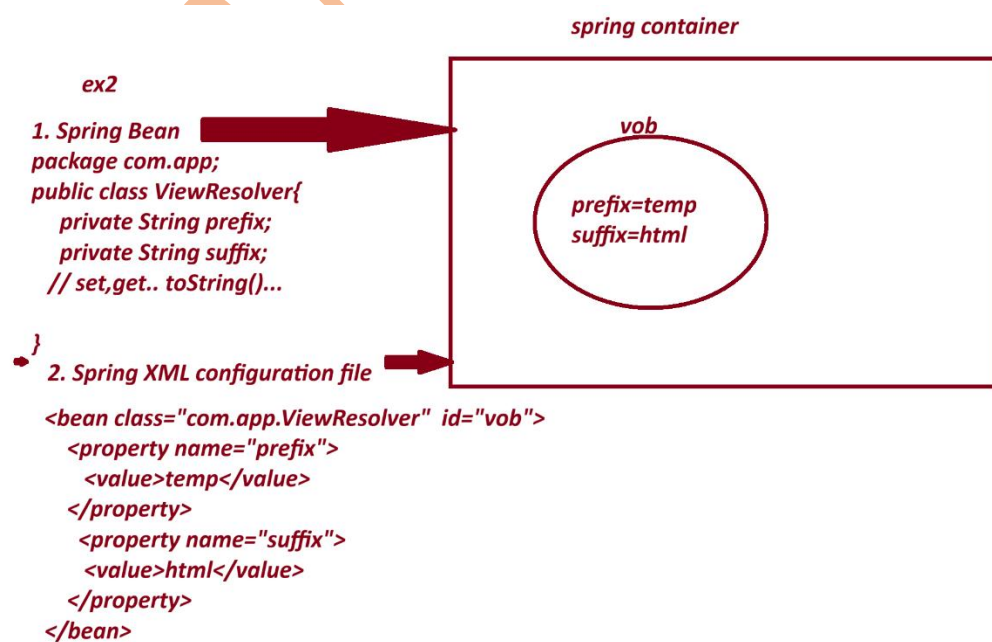
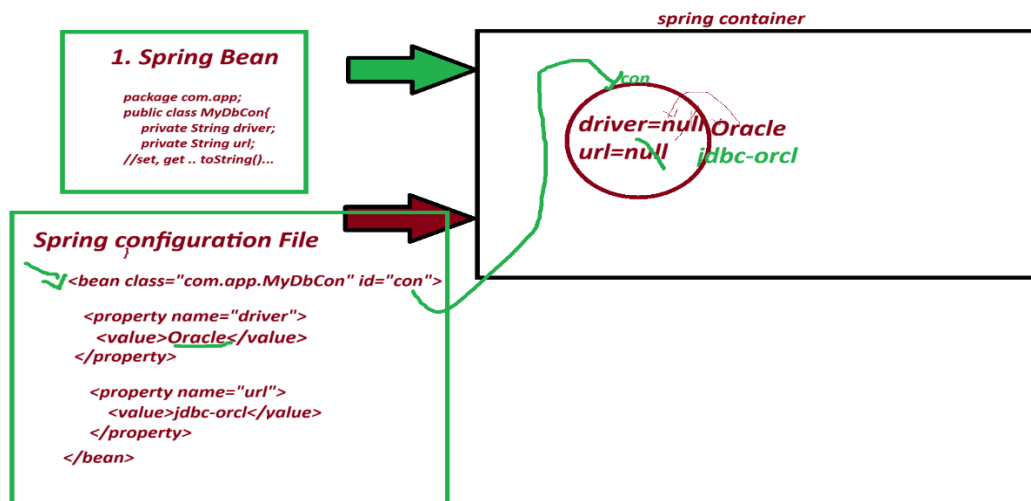
<bean id="objName" class="FullyQualifiedClassName"></bean>

2. To call set method

<property name="variableName"></property>

3. To provide data to Primitive Variables

<value>data</value>



1. Spring bean

package com.app;

```
public class PaymentService{  
  
    private int token;  
  
    private boolean active;  
  
    private String provider;  
  
    //set.. set toString..etc  
  
}
```

2. Spring XML configuration file.

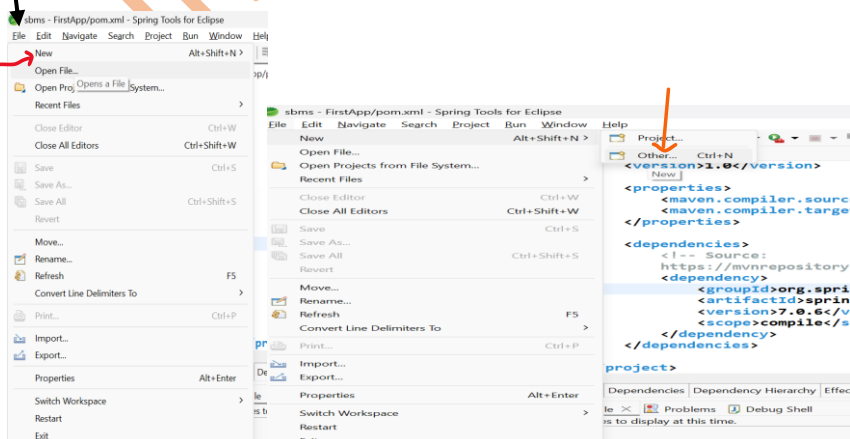
```
<bean class="com.app.PaymentService" id="psObj">  
  
    <property name="token">  
  
        <value>154678</value>  
  
    </property>  
  
    <property name="active">  
  
        <value>true</value>  
  
    </property>  
  
    <property name="provider">  
  
        <value>xyzBank</value>  
  
    </property>  
  
</bean>
```

=====

CREATE MAVEN PROJECT WITH BELOW STEP :-

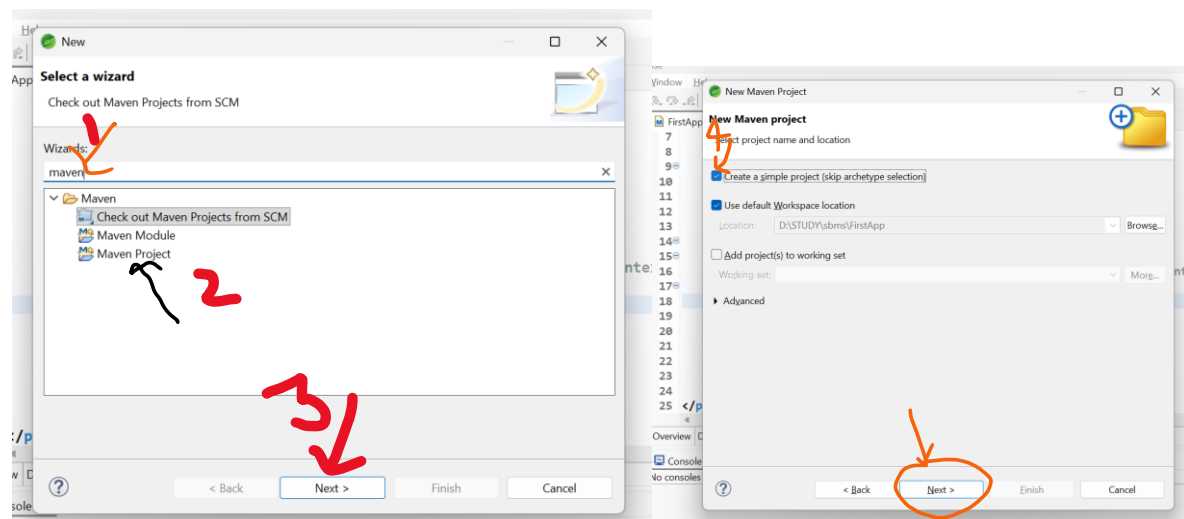
=====

1. Create Project



File-> New-> Other-> Type maven->choose maven project-> next->

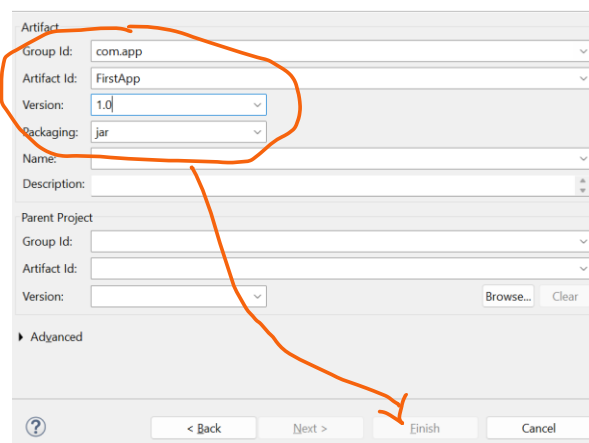
choose checkbox [v] create simple project -> next-> Enter details



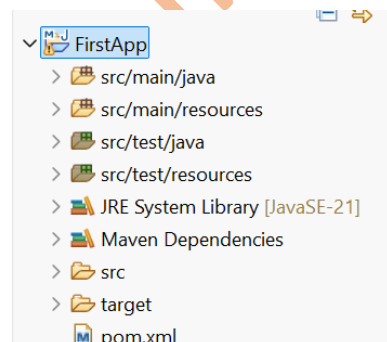
groupId : com.app

ArtifactId : FirstApp

version : 1.0



-> Finish



2. pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>FirstApp</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>

<dependencies>

    <!-- Source:
        https://mvnrepository.com/artifact/org.springframework/spring-context -->

    <dependency>

        <groupId>org.springframework</groupId>

        <artifactId>spring-context</artifactId>

        <version>7.0.6</version>

        <scope>compile</scope>

    </dependency>

</dependencies>

</project>

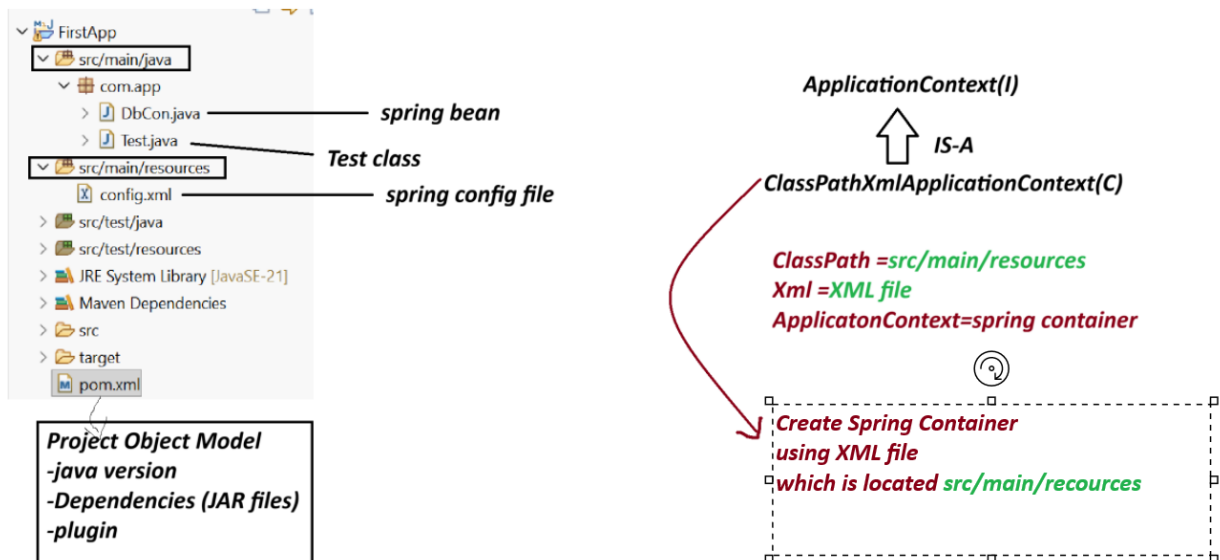
```

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Spring Core First Application 3 Files.

1. Spring bean (.java)
2. Spring XML configuration (config.xml)
3. Test class(.java)



Note:-

1. Spring container are two Types.

(a) Legacy Container : BeanFactory(I) support only XML

(b) New Container : ApplicationContext(i) support all XML/java/Annotation config.

(c) ApplicationContext(I) : it is a interface.

we are using one of its Impl class:

ClassPathXmlApplicationContext(C).

=====

Create Maven Project with below Step:

=====

File>new>other>type maven>choose maven project>next>choose checkbox> create Simple project>next>Enter details

GroupId - com.app

ArtificatId - SecondApp

version - 1.0

code

SecondApp

=====

pom.xml

=====

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>
<artifactId>SecondApp</artifactId>
<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>

<dependencies>
<!-- Source:https://mvnrepository.com/artifact/org.springframework/spring-context -->
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
</project>
```

DbCon.java

```
package com.app;

public class DbCon {
    private String driver;
    private String url;
```

```

//alt+shit+s

public DbCon() {
super();
System.out.println("Constructor called");
}

public String getDriver() {
return driver;
}

public void setDriver(String driver) {
this.driver = driver;
System.out.println("Driver method called");
}

public String getUrl() {
return url;
}

public void setUrl(String url) {
this.url = url;
System.out.println("Url method called");
}

@Override
public String toString() {
return "DbCon [driver=" + driver + ", url=" + url + "];"
}

}

```

=====

config.xml

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

```

```
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean class="com.app.DbCon" id="mobj">
    <property name="driver">
        <value>Oracle Driver</value>
    </property>

    <property name="url">
        <value>JDBC-ORCL</value>
    </property>
</bean>
```

```
</beans>
```

```
=====
```

Test.java

```
-----
```

```
package com.app;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
public class Test {
```

```
//ctrl+shit+O
```

```
public static void main(String[] args) {
```

```
    ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
```

```
    Object ob = ac.getBean("mobj");
```

```
    System.out.println(ob);
```

```
}
```

```
}
```

output

Constructor called

Driver method called

Url method called

DbCon [driver=Oracle Driver, url=JDBC-ORCL]

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Spring Container (DI - Depedency Injection)

=>Depedency - Variables exist in class

3 type

1. Primitive Type (8+1) : byte, short, int, long, float, double, boolean, char, String

2. Collection Type (4) : Set, List, Map and Properties

3. Reference Type (x) : using a class/interface variable

=> Injection : provide Data

- Setter Injection

- constructor Injection

LookUpMethod Injetion

- Field Injection (Annotations)

=====

XML TAG

=====

<bean> - Object creation

<property> - calling set method

<constructor-arg> - call parameterized constructor

<value> - To provide data to Primitive Type.

<list> <set> <map> <props> -- To provide data to Collection type.

<ref/> -- To Link two objects

=====

Collection Configuration

=====

=> Spring container support working with collection configuration which are from java.util package

=====

Collection Name	Collection Type	XML Tag Name
=====	=====	=====
List	Interface	<list>
Set	Interface	<set>
Map	Interface	<map> and <entry>
Properties	Class	<props> and <prop>

List -> ArrayList

Set -> LinkedHashSet

Map -> LinkedHashMap

Properties --> NA

coding part collectionTypeS8

pom.xml

<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>collectionTypeS8</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>

```
</properties>
```

```
<dependencies><!-- Source:
```

```
https://mvnrepository.com/artifact/org.springframework/spring-context -->
```

```
<dependency>
```

```
<groupId>org.springframework</groupId>
```

```
<artifactId>spring-context</artifactId>
```

```
<version>7.0.5</version>
```

```
<scope>compile</scope>
```

```
</dependency>
```

```
</dependencies>
```

```
</project>
```

```
-----  
Spring bean
```

```
MyDetails.java  
-----
```

```
package com.app;
```

```
import java.util.List;
```

```
import java.util.Map;
```

```
import java.util.Properties;
```

```
import java.util.Set;
```

```
public class MyDetails {
```

```
//ctrl+shift+O
```

```
private List<String> data;
```

```
private Set<String> models;
```

```
private Map<Integer,String> design;
```

```
private Properties myinfo;
```

```
public MyDetails() {
```

```
super();
```

```
}
```

```
public List<String> getData() {  
    return data;  
}
```

```
public void setData(List<String> data) {  
    this.data = data;  
}
```

```
public Set<String> getModels() {  
    return models;  
}
```

```
public void setModels(Set<String> models) {  
    this.models = models;  
}
```

```
public Map<Integer, String> getDesign() {  
    return design;  
}
```

```
public void setDesign(Map<Integer, String> design) {  
    this.design = design;  
}
```

```
public Properties getMyinfo() {  
    return myinfo;  
}  
public void setMyinfo(Properties myinfo) {  
    this.myinfo = myinfo;  
}
```

```
@Override
```

```
public String toString() {
```

```
return "MyDetails [data=" + data + ", models=" + models + ", design=" + design + ", myinfo=" + myinfo +
    "];"
}
}
```

xml config file

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
```

```
xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean class="com.app.MyDetails" id="myinfo">
```

```
<property name="data">
```

```
<list>
```

```
<value>A</value>
```

```
<value>B</value>
```

```
<value>C</value>
```

```
<value>A</value>
```

```
</list>
```

```
</property>
```

```
<property name="models">
```

```
<set>
```

```
<value>M1</value>
```

```
<value>M2</value>
```

```
<value>M3</value>
```

```
<value>M1</value>
```

```
</set>
```

```
</property>
```

<property name="design">

<map>

<entry>

<key>

<value>101</value>

</key>

<value>AB</value>

</entry>

<entry>

<key>

<value>102</value>

</key>

<value>Suraj</value>

</entry>

<entry>

<key>

<value>103</value>

</key>

<null></null>

</entry>

</map>

</property>

<property name="myinfo">

<props>

<prop key="A1">B1</prop>

<prop key="A2">B2</prop>

<prop key="A3">B3</prop>

<prop key="A5"></prop>

</props>

```
</property>
```

```
</bean>
```

```
</beans>
```

Test File

Test.java

```
package com.app;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
public class Test {
```

```
    public static void main(String[] args) {
```

```
        ApplicationContext ac=new ClassPathXmlApplicationContext("config.xml");
```

```
        //Object ob = ac.getBean("myinfo");
```

```
        MyDetails ob = ac.getBean("myinfo",MyDetails.class);
```

```
        System.out.println(ob);
```

```
        System.out.println(ob.getData().getClass().getName());
```

```
        System.out.println(ob.getModels().getClass().getName());
```

```
        System.out.println(ob.getDesign().getClass().getName());
```

```
        System.out.println(ob.getMyinfo().getClass().getName());
```

```
    }
```

```
}
```

console output

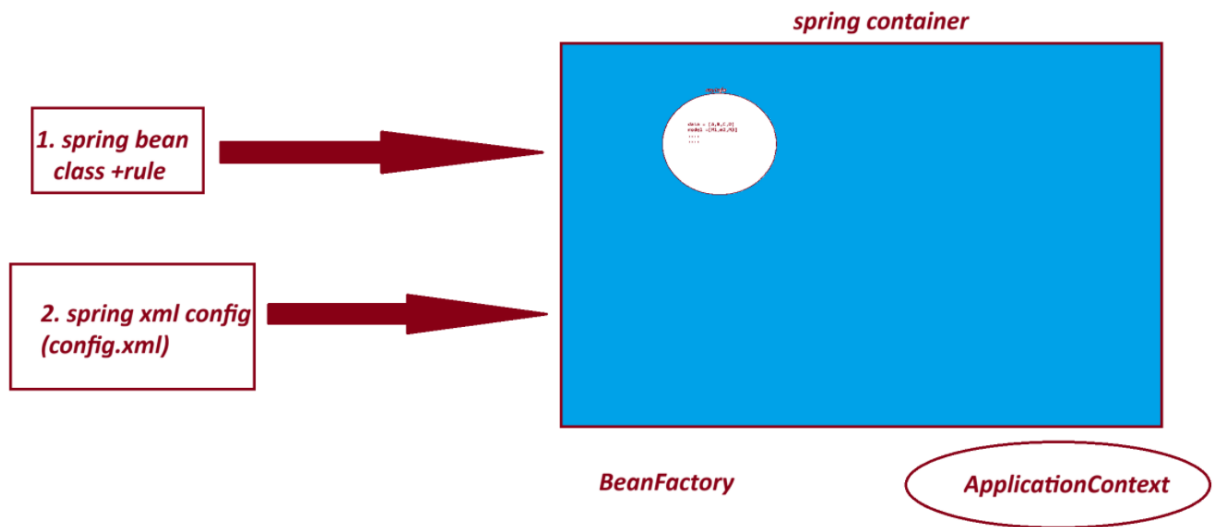
```
MyDetails [data=[A, B, C, A], models=[M1, M2, M3], design={101=AB, 102=Suraj, 103=null},  
myinfo={A1=B1, A2=B2, A3=B3, A5=}]
```

```
java.util.ArrayList
```

```
java.util.LinkedHashSet
```

java.util.LinkedHashMap

java.util.Properties



===

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3. Reference Type Dependency - Setter Injection.

=====

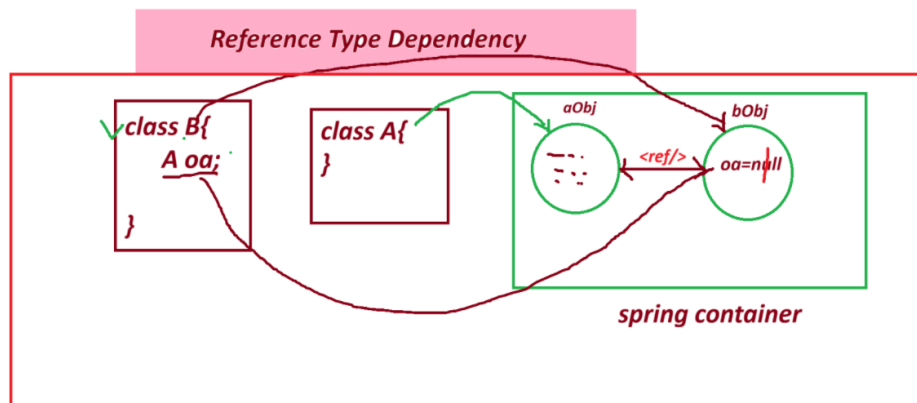
=> <ref/> : - tag is used to Link objects in case of HAS-A Relation.

Syntax:

<property name="variableName">

<ref bean="childObjName"/>

</property/>



=====

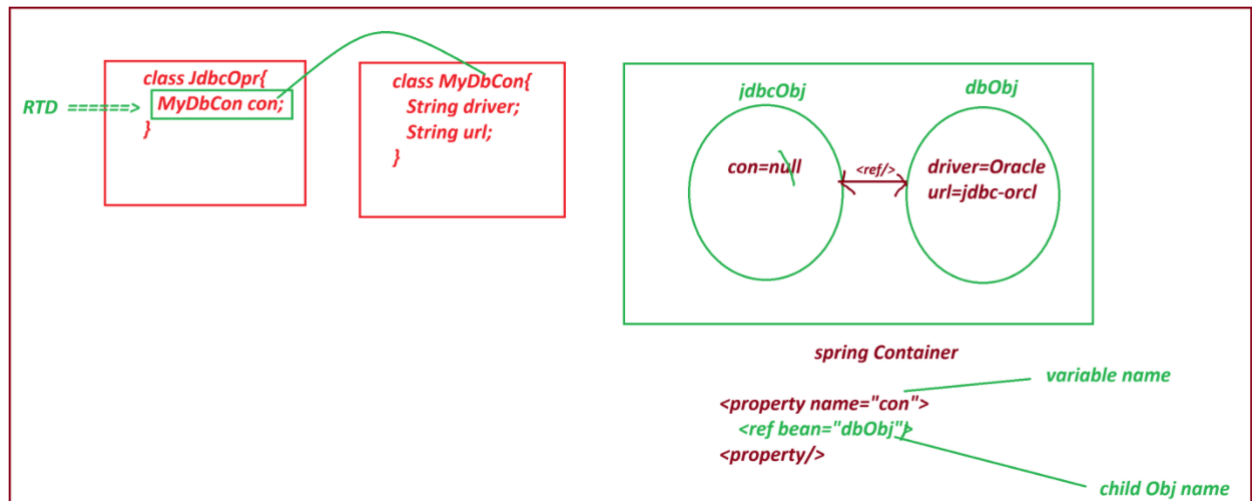
- If we do not use <ref/> then ref variable holds default value null.

=> spring container creates object to both classes and link them by using <ref/> tag.

=> if child object nam (ref bean name) is not valid or not exist then spring container throws exception like

NoSuchBeanDefinitionException:

=====



project : ReferenceTypeS9

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.app</groupId>
<artifactId>ReferenceTypeS9</artifactId>
<version>1.0</version>
<properties>
<maven.compiler.source>21</maven.compiler.source>
<maven.compiler.target>21</maven.compiler.target>
</properties>
<dependencies>
```

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>7.0.5</version>
<scope>compile</scope>
</dependency>
</dependencies>
```

```
</project>
```

```
-----
JdbcOpr.java
-----
```

```
package com.app;
```

```
public class JdbcOpr {
    private MyDbCon con; //RTD

    public JdbcOpr() {
        super();
        // TODO Auto-generated constructor stub
    }

    public MyDbCon getCon() {
        return con;
    }

    public void setCon(MyDbCon con) {
        this.con = con;
    }
}
```

```
@Override  
public String toString() {  
return "JdbcOpr [con=" + con + "];"  
}
```

```
}
```

```
=====
```

MyDbCon.java

```
-----
```

```
package com.app;
```

```
public class MyDbCon {
```

```
    private String driver;
```

```
    private String url;
```

```
    public MyDbCon() {
```

```
        super();
```

```
        // TODO Auto-generated constructor stub
```

```
    }
```

```
    public String getDriver() {
```

```
        return driver;
```

```
    }
```

```
    public void setDriver(String driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    public String getUrl() {
```

```
        return url;
```

```
    }
```

```

    public void setUrl(String url) {
this.url = url;
    }

    @Override
    public String toString() {
return "MyDbCon [driver=" + driver + ", url=" + url + "];"
    }

    //alt+shift+s

}

```

config.xml

```

-----
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean class="com.app.MyDbCon" id="dbObj" >
        <property name="driver">
            <value>OracleDriver</value>
        </property>
        <property name="url">
            <value>JDBC-ORCL</value>
        </property>
    </bean>

    <bean class="com.app.JdbcOpr" id="jdbcObj">
        <property name="con">
            <ref bean="dbObj"/>

```

</property>

</bean>

</beans>

Test.java

package com.app;

import org.springframework.context.ApplicationContext;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Test {

public static void main(String[] args) {

ApplicationContext ac = new ClassPathXmlApplicationContext("config.xml");

JdbcOpr jdbc = ac.getBean("jdbcObj", JdbcOpr.class);

System.out.println(jdbc);

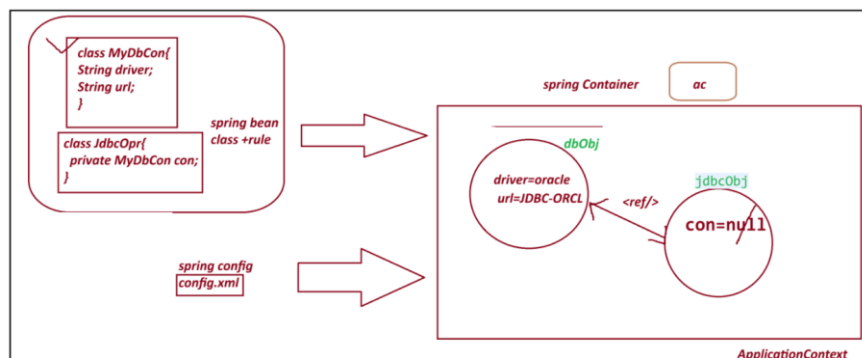
}

}

output

JdbcOpr [con=MyDbCon [driver=OracleDriver, url=JDBC-ORCL]]

=====



SI - Setter Injection :

Spring container creates object to given class using default constructor and provide data using setter method.

Employee emp=new Employee(); =====> <bean id="emp" class="Employee"/>

emp.setEmpId(10); =====> <property name="empId" value="10"/>

=====

*** Constructor Injection CI *****

=====

=> Spring container creating object and providing data using Parameterized constructor.

Here , we use <constructor-arg> tag to indicate.

Employee emp=new Employee(10, "A"); =====>

<bean>

<constructor-arg>

<value>10</value>

</constructor-arg>

<constructor-arg>

<value>A</value>

</constructor-arg>

</bean>

Note:

A. If a class has 3 param constructor then we need to pass <constructor-arg> 3 times in xml config file.

B. Data must be passed in order using <constructor-arg> as per constructor defined in class.

=====

code for CI (constructor Injection)

pom.xml

```

-----
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.app</groupId>

<artifactId>ConstructorInjectionS9</artifactId>

<version>1.0</version>

<properties>

<maven.compiler.source>21</maven.compiler.source>

<maven.compiler.target>21</maven.compiler.target>

</properties>

<dependencies>

<dependency>

<groupId>org.springframework</groupId>

<artifactId>spring-context</artifactId>

<version>7.0.5</version>

<scope>compile</scope>

</dependency>

</dependencies>

</project>

```

```

-----
Product.java

```

```

package com.app;

```

```

import java.util.List;

```

```

public class Product {

```

```

    private String pcode;

```

```

private String model;

private List<String> data;

private Vendor vob;


public Product(String pcode, String model, List<String> data, Vendor vob) {
    super();
    this.pcode = pcode;
    this.model = model;
    this.data = data;
    this.vob = vob;
}

@Override
public String toString() {
    return "Product [pcode=" + pcode + ", model=" + model + ", data=" + data + ", vob=" + vob + "];"
}

}

```

Test.java

```

package com.app;


public class Vendor {

    private String vcode;
    private String vaddr;


    public Vendor(String vcode, String vaddr) {
        super();
        this.vcode = vcode;
        this.vaddr = vaddr;
    }
}

```


@Override

```
public String toString() {  
    return "Vendor [vcode=" + vcode + ", vaddr=" + vaddr + "];"  
}
```

```
}
```

config.xml

```
<?xml version="1.0" encoding="UTF-8"?>  
  
<beans xmlns="http://www.springframework.org/schema/beans"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="  
        http://www.springframework.org/schema/beans  
        http://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean class="com.app.Vendor" id="v1">
```

```
    <constructor-arg>
```

```
        <value>v1109</value>
```

```
    </constructor-arg>
```

```
    <constructor-arg>
```

```
        <value>IND</value>
```

```
    </constructor-arg>
```

```
</bean>
```

```
<bean class="com.app.Product" id="pobj">
```

```
    <constructor-arg>
```

```
        <value>P01</value>
```

```
    </constructor-arg>
```

```
    <constructor-arg>
```

<value>HP-23444</value>

</constructor-arg>

<constructor-arg>

<list>

<value>AA</value>

<value>BB</value>

</list>

</constructor-arg>

<constructor-arg>

<ref bean="v1"/>

</constructor-arg>

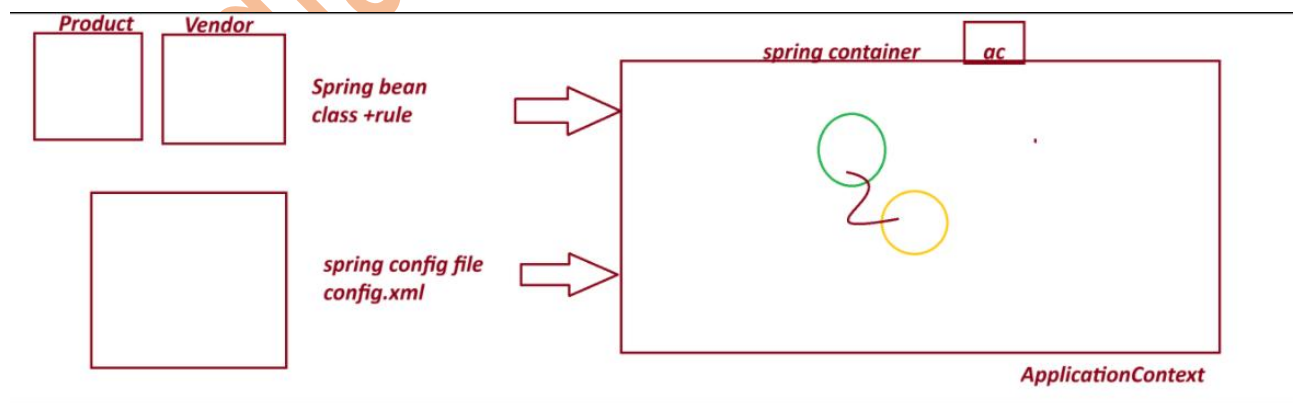
</bean>

</beans>

output

=====

Product [pcode=P01, model=HP-23444, data=[AA, BB], vob=Vendor [vcode=v1109, vaddr=IND]]



=====

session 10

=====

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